MOR – 2025 – (MS)^{2}SC

MOR practical session
Session 2: solving the unsteady heat equation with PGD (a-priori)

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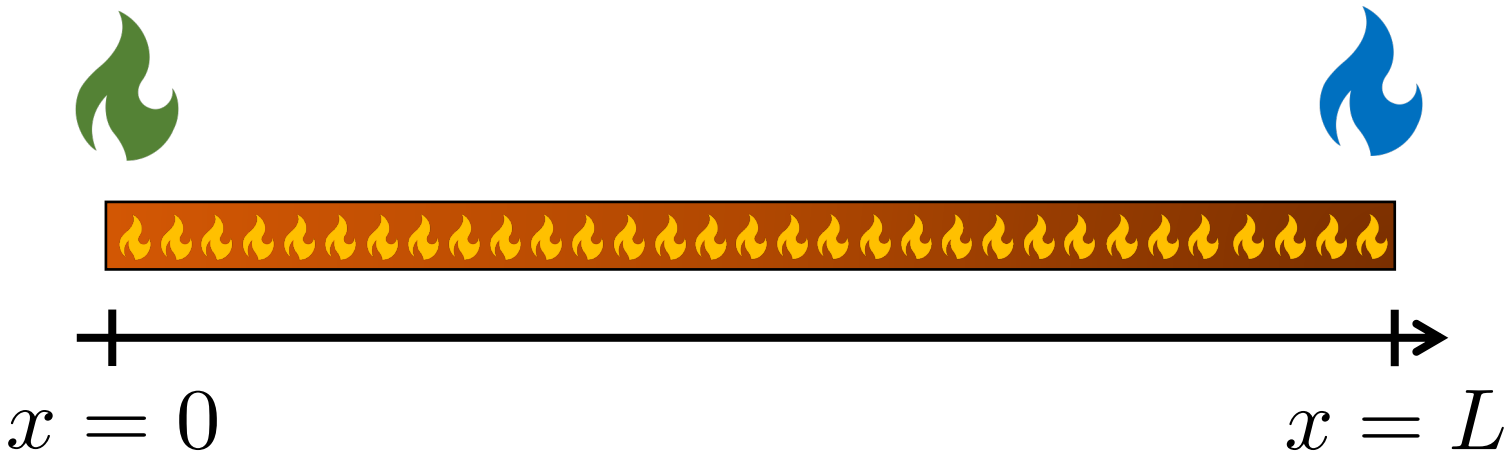
Our problem

Unsteady heat equation:

$$
(\mathrm{E}) : \begin{cases} \rho c \dfrac{\partial \boldsymbol{u}}{\partial t} - k \dfrac{\partial^2 \boldsymbol{u}}{\partial x^2} = \boldsymbol{f} \\ \boldsymbol{u}_d^0 = \boldsymbol{u}(0,t) = \sin(\frac{2\pi}{T} t) \\ \boldsymbol{u}_d^L = \boldsymbol{u}(L,t) = -\sin(\frac{4\pi}{T} t) \\ \boldsymbol{u}(\boldsymbol{x},0) = \boldsymbol{0} \end{cases}
$$



$$t \in [0,T] = \boldsymbol{I}$$

Definition of spaces

Intervals:

- $\boldsymbol{X} \in \boldsymbol{\Omega}$, a non-empty bounded domain, where $\boldsymbol{\Omega} \subset \mathbb{R}^d$ $(d = 1, 2, 3)$
- $\boldsymbol{I} = [0, T]$ associated with the temporal variable t; we also define: $\| \bullet \|_{\boldsymbol{I}} = \left(\int_{\boldsymbol{I}} \bullet^2 \, \mathrm{d}t\right)^{1/2}$

Vector spaces:

- $\mathcal{U} \equiv \mathcal{H}^1(\boldsymbol{\Omega}) = \left\{\boldsymbol{u} \in \boldsymbol{L}^2(\boldsymbol{\Omega}), \nabla \boldsymbol{u} \in \boldsymbol{L}^2(\boldsymbol{\Omega}) \mid \boldsymbol{u}_{|\partial_u \Omega} = \boldsymbol{U}_d\right\}$, with $\| \bullet \|_{\mathcal{U}} = \left(\int_{\boldsymbol{\Omega}} \bullet^2 \, \mathrm{d}\boldsymbol{\Omega}\right)^{1/2}$ as the $\boldsymbol{L}^2$ norm of $\mathcal{U}$
- $\mathcal{U}_0 = \mathcal{H}^1_0(\boldsymbol{\Omega}) = \left\{\boldsymbol{u} \in \mathcal{H}^1(\Omega) \mid \boldsymbol{u}_{|\partial_u \Omega} = \boldsymbol{0}\right\}$
- $\mathcal{I} \equiv \boldsymbol{L}^2(\boldsymbol{I}, \mathbb{R})$ with $\boldsymbol{\lambda}_{|t=0} = \boldsymbol{\lambda}(0)$ and $\dot{\boldsymbol{\lambda}}_{|t=0} = \dot{\boldsymbol{\lambda}}(0)$
- $\mathcal{W} \equiv \boldsymbol{L}^2(\boldsymbol{I}, \mathcal{U}) = \left\{\boldsymbol{u} : \boldsymbol{\Omega} \times \boldsymbol{I} \to \mathbb{R}^3; \forall t, \boldsymbol{X} \mapsto \boldsymbol{u}(\boldsymbol{X}, t) \mid \|\boldsymbol{u}(\boldsymbol{X}, t)\|_{\mathcal{W}} < +\infty, \boldsymbol{u}(\boldsymbol{X}, t)_{|\partial_u \Omega} = \boldsymbol{U}_d(\boldsymbol{X}, t), \boldsymbol{u}(\boldsymbol{X}, t)_{|t=0} = \boldsymbol{u}_0, \dot{\boldsymbol{u}}(\boldsymbol{X}, t)_{|t=0} = \dot{\boldsymbol{u}}_0\right\}$
  where we define: $\| \bullet \|_{\mathcal{W}} = \left(\int_{\boldsymbol{\Omega} \times \boldsymbol{I}} \bullet^2 \, \mathrm{d}\boldsymbol{\Omega} \, \mathrm{d}t\right)^{1/2}$
- $\mathcal{W}_0 \equiv \boldsymbol{L}^2(\boldsymbol{I}, \mathcal{U}) = \left\{\boldsymbol{u} : \boldsymbol{\Omega} \times \boldsymbol{I} \to \mathbb{R}^3; \forall t, \boldsymbol{X} \mapsto \boldsymbol{u}(\boldsymbol{X}, t) \mid \|\boldsymbol{u}(\boldsymbol{X}, t)\|_{\mathcal{W}} < +\infty, \boldsymbol{u}(\boldsymbol{X}, t)_{|\partial_u \Omega} = \boldsymbol{0}, \boldsymbol{u}(\boldsymbol{X}, t)_{|t=0} = \boldsymbol{u}_0, \dot{\boldsymbol{u}}(\boldsymbol{X}, t)_{|t=0} = \dot{\boldsymbol{u}}_0\right\}$
- $\tilde{\mathcal{W}} \equiv \mathcal{U}_0 \otimes \mathcal{I}$

Sub - vector spaces:

- U a sub-vector space of $\mathcal{U}$ of finite dimension N_x
- U_0 a sub-vector space of $\mathcal{U}_0$ of finite dimension N_x

First step: a reference solution

1. Variational formulation (weak form) in space

2. Semi-discretization in space: finite elements of order 1
   - construction of finite-element operators

3. Time discretization and integration scheme
   - Euler scheme

4. Boundary conditions taken into account:
   - by substitution method
   - by a change of variable

1- Variational formulation

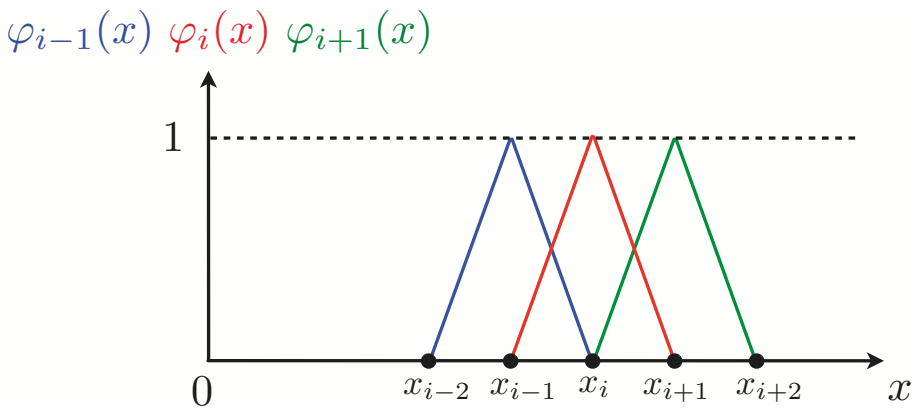
$$(\mathrm{E}) : \begin{cases} \rho c \dfrac{\partial \boldsymbol{u}}{\partial t} - k \dfrac{\partial^2 \boldsymbol{u}}{\partial x^2} = \boldsymbol{f} \\ \boldsymbol{u}_d^0 = \boldsymbol{u}(0,t) = \sin(\frac{2\pi}{T} t) \\ \boldsymbol{u}_d^L = \boldsymbol{u}(L,t) = -\sin(\frac{4\pi}{T} t) \\ \boldsymbol{u}(\boldsymbol{x},0) = \boldsymbol{0} \end{cases}$$

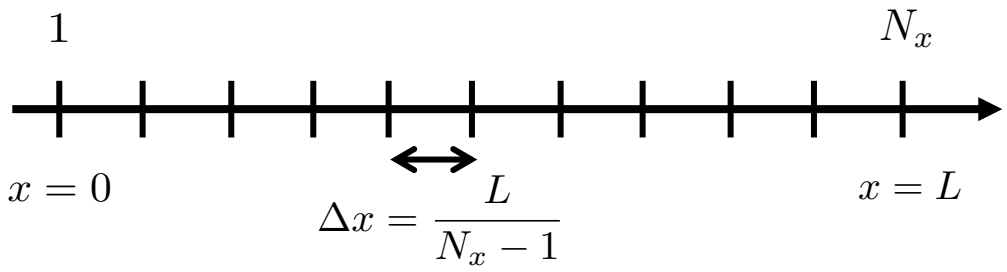
$$\vdots$$

$$\forall t, \forall \boldsymbol{v} \in \mathcal{U}_0, \int_0^L \rho c \dot{\boldsymbol{u}}\, \boldsymbol{v} \mathrm{dx} + \int_0^L k \frac{\partial \boldsymbol{u}}{\partial x} \frac{\partial \boldsymbol{v}}{\partial x} \mathrm{dx} = \int_0^L \boldsymbol{f}\, \boldsymbol{v} \mathrm{dx}$$

2- Semi-discretization in space

$$\forall t, \forall \boldsymbol{v} \in \mathcal{U}_0, \int_0^L \rho c \dot{\boldsymbol{u}}\, \boldsymbol{v} \mathrm{dx} + \int_0^L k \frac{\partial \boldsymbol{u}}{\partial x} \frac{\partial \boldsymbol{v}}{\partial x} \mathrm{dx} = \int_0^L \boldsymbol{f}\, \boldsymbol{v} \mathrm{dx}$$





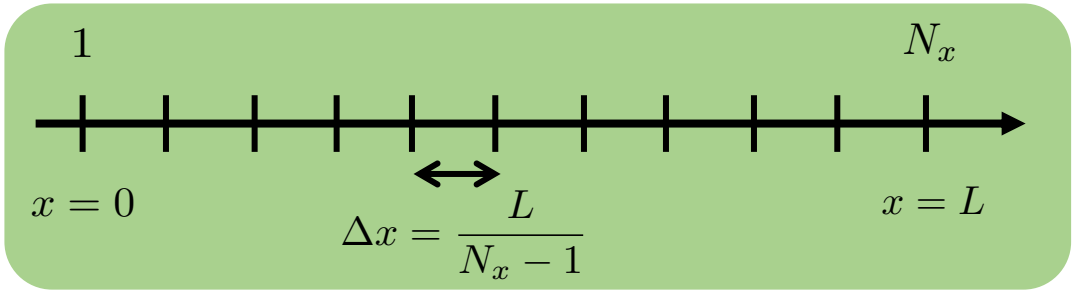
$\mathrm{U}_0 = \mathrm{Span}\left(\{\boldsymbol{\varphi}_i\}_{i \in \{1,\ldots,N_x\}}\right)$ a finite-dimensional sub-vector space of $\mathcal{U}_0$

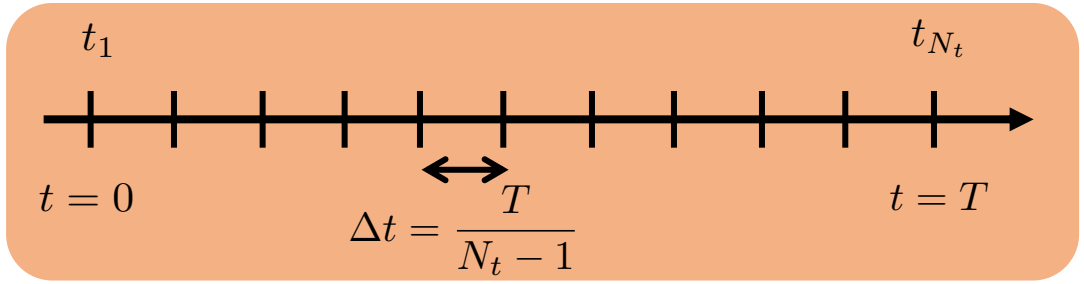
$$\forall t, \forall \underline{V} \in \mathrm{U}_0, \quad \underline{V}^\top \left[\mathbb{M}\, \underline{\dot{U}}(t) + \mathbb{K}\, \underline{U}(t) - \underline{F}(t)\right] = 0$$

$$\vdots$$

construction of finite-element operators: $\mathbb{M} = ?$ $\mathbb{K} = ?$

3- Time discretization





Notations used:

$$\underline{\underline{U}}\ (N_x \times N_t)$$
$$\underline{\underline{F}}\ (N_x \times N_t)$$

$$\underline{\underline{U}} = \begin{pmatrix} | & & | & & | \\ \underline{U}(t_1) & \cdots & \underline{U}(t_i) & \cdots & \underline{U}(t_{N_t}) \\ | & & | & & | \end{pmatrix}$$

N_x

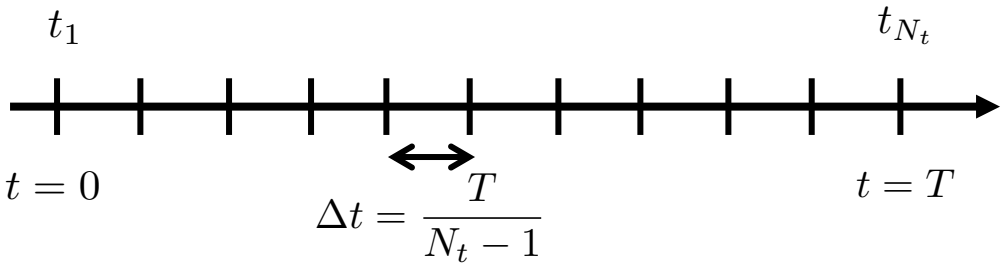
N_t

$$\int_0^L \boldsymbol{f}\, \boldsymbol{v} \mathrm{d}x \longrightarrow \underline{v}^\top \underline{F}(t) = ? \qquad \underline{\underline{F}} = ?$$

$$\underline{\underline{f}}\ (N_x \times N_t)$$

space-time discretization of the volume loading field

3- Time discretization



$$\underline{V}^\top \left[\mathbb{M}\, \dot{\underline{U}}(t) + \mathbb{K}\, \underline{U}(t) - \underline{F}(t) \right] = 0$$

$$\mathbb{M}\, \dot{\underline{U}}(t_i) + \mathbb{K}\, \underline{U}(t_i) - \underline{F}(t_i) = \underline{0}$$

Retrograde Euler scheme (stable/ implicit) :

$\forall i \in \{2, ..., N_t\}$,

$$\mathbb{M}\, \frac{\underline{U}(t_i) - \underline{U}(t_{i-1})}{\Delta t} + \mathbb{K}\, \underline{U}(t_i) - \underline{F}(t_i) = \underline{0}$$

$$\underbrace{\left(\frac{\mathbb{M}}{\Delta t} + \mathbb{K} \right)}_{\mathbb{H}} \underline{U}(t_i) - \underbrace{\left(\underline{F}(t_i) + \frac{\mathbb{M}}{\Delta t} \underline{U}(t_{i-1}) \right)}_{\underline{G}(t_i)} = \underline{0}$$

4- Boundary conditions

unknown

known because imposed

test function: our choice

$$\underline{V}^\top \left[\mathbb{H}\underline{U}(t_i) - \underline{G}(t_i)\right] = 0$$

Substitution method: [Boucard, Structural assemblies: Dealing with frictional contact problems ; (MS)²SC lesson]

(reorganization of degrees of freedom in the system)

$$\begin{pmatrix} \underline{V}_u \\ \underline{0} \end{pmatrix}^\top \left[\begin{pmatrix} \mathbb{H}_{uu} & \mathbb{H}_{ud} \\ \mathbb{H}_{du} & \mathbb{H}_{dd} \end{pmatrix} \begin{pmatrix} \underline{U}_u(t_i) \\ \underline{U}_d(t_i) \end{pmatrix} - \begin{pmatrix} \underline{G}_u(t_i) \\ \underline{G}_d(t_i) \end{pmatrix} \right] = 0$$

$$\underline{U}_u(t_i) = \mathbb{H}_{uu}^{-1} \left(\underline{G}_u(t_i) - \mathbb{H}_{ud}\underline{U}_d(t_i)\right)$$

Knowing the free and fixed degrees of freedom, how can you solve this new problem in Matlab?

Solve the problem.

4- Boundary conditions

$$\boldsymbol{u}(\boldsymbol{X},t) = \underbrace{\boldsymbol{u}_{\mathrm{CL}}(\boldsymbol{X},t)}_{\in\mathcal{W}} + \underbrace{\boldsymbol{w}(\boldsymbol{X},t)}_{\in\mathcal{W}_0}$$

$$\underline{\underline{U}} = \underbrace{\underline{\underline{U}}_{\mathrm{CL}}}_{\text{DA}} + \underbrace{\underline{\underline{W}}}_{\text{DA to 0}}$$

Change of variable (« relèvement »):

$$\underline{V}^\top \left[\mathbb{H}\underline{U}(t_i) - \underline{G}(t_i)\right] = 0$$

« **D**ynamically **A**dmissible »: complies perfectly with boundary conditions and initial conditions and is sufficiently regular over the rest of the domain.

$\underline{U}_{\mathrm{CL}}(t) \in \mathrm{U}$ and $\underline{U}_{\mathrm{CL}}(0) = \underline{0}$

$$\underline{V}^\top \left[\mathbb{H}\underline{W}(t_i) - \underbrace{\left(\underline{G}(t_i) - \mathbb{H}\underline{U}_{\mathrm{CL}}(t_i)\right)}_{\underline{J}(t_i)}\right] = 0$$

Substitution : $$\underline{W}_u(t_i) = \mathbb{H}_{uu}^{-1}\left(\underline{J}_u(t_i) - \mathbb{H}_{ud}\cancel{\underline{W}_d}(t_i)\right) = \mathbb{H}_{uu}^{-1}\underline{J}_u(t_i)$$

$$\underline{W}_d(t_i) = \underline{0}$$

For example, for our problem, what can we use as the $\underline{\underline{U}}_{\mathrm{CL}}$ field?

We can draw inspiration from first-order shape functions...

4- Boundary conditions

Back to the original discretized equation with:

$$\underline{\underline{U}} = \underbrace{\underline{\underline{U}}_{\text{CL}}}_{\text{DA}} + \underbrace{\underline{\underline{W}}}_{\text{DA to 0}}$$

$$\underline{V}^{\top}\left[\mathbb{M}\,\underline{\dot{W}}(t_i) + \mathbb{K}\,\underline{W}(t_i) - \underbrace{\left(\underline{R}(t_i) - \mathbb{M}\,\underline{\dot{U}}_{CL}(t_i) - \mathbb{K}\,\underline{U}_{CL}(t_i)\right)}_{\underline{Q}(t_i)}\right] = 0$$

$$\underline{\underline{\dot{U}}}_{CL} = \underline{\underline{U}}_{CL}\,\mathbb{D}_t$$

$\mathbb{D}_t = ?$ *must be consistent with our integration scheme! (**retrograde EULER**)*
(what to do for the first time step?)

Solve the problem with the change of variable.

Second step: resolution by PGD

1. Writing the problem
   - Time Galerkin vision
   - Fixed-point equations

2. Implementing the PGD solver
   - Code tips and tricks

3. Improving the algorithm
   - Orthogonalization of modes (see previous session)
   - Spatial mode update phase (POD on space mode)

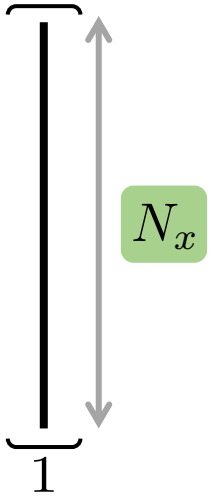
4. Comparaison with SVD decomposition?

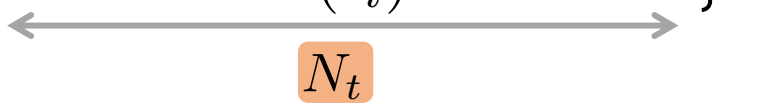
I- Writing the problem

$\boldsymbol{w} \in \tilde{\mathcal{W}}$ $\quad \underline{\Lambda} \in \mathrm{U}_0$

$$\underline{U}(t) = \underline{U}_{\mathrm{CL}}(t) + \underbrace{\underline{U}_{i-1}(t) + \underline{\Lambda}\,\lambda(t)}_{\underline{W}(t)} \qquad \underline{U}_{i-1}(t) = \sum_{k=0}^{i-1} \underline{\Lambda}_k\,\lambda_k(t)\ ,\ \underline{\Lambda}_k \in \mathrm{U}_0$$

Notation used: $\underline{\Lambda}\,(N_x \times 1)$ $\qquad \underline{\lambda}\,(1 \times N_t)$

$\underline{\Lambda} =$ 

$\underline{\lambda} =$ 

$\lambda(t) \in \mathbb{R}$

I- Writing the problem

$$\underline{U}(t) = \underline{U}_{\mathrm{CL}}(t) + \underbrace{\underline{W}_{i-1}(t) + \underline{\Lambda}\,\lambda(t)}_{\underline{W}(t)} \qquad \underline{\Lambda} \in \mathrm{U}_0 \qquad \underline{W}_{i-1}(t) = \sum_{k=0}^{i-1} \underline{\Lambda}_k\,\lambda_k(t)$$

1. Weak time form on discretized equation : [Ladevèze, 1985][Ladevèze, 1999] [Chinesta, 2011]...

$$\forall \lambda^*(t), \forall \underline{\Lambda}^* \in \mathrm{U}_0$$

$$\int_0^T \left(\underline{\Lambda}\,\lambda^*(t) + \underline{\Lambda}^*\,\lambda(t)\right)^\top \left[\mathbb{M}\left(\dot{\underline{U}}_{\mathrm{CL}}(t) + \dot{\underline{W}}_{i-1}(t) + \underline{\Lambda}\,\dot{\lambda}(t)\right) + \mathbb{K}\left(\underline{U}_{\mathrm{CL}}(t) + \underline{W}_{i-1}(t) + \underline{\Lambda}\,\lambda(t)\right) - \underline{F}(t)\right] \mathrm{dt} = 0$$

$$\vdots$$

$$\int_0^T \left(\underline{\Lambda}\,\lambda^*(t) + \underline{\Lambda}^*\,\lambda(t)\right)^\top \left[\mathbb{M}\underline{\Lambda}\,\dot{\lambda}(t) + \mathbb{K}\underline{\Lambda}\,\lambda(t) - \underline{Q}(t)\right] \mathrm{dt} = 0$$

$$\vdots$$

$$\underline{Q}_0(t) = ? \qquad \underline{Q}(t) = ?$$

For i=1, first mode generation

I- Writing the problem

$\forall \lambda^*(t), \forall \underline{\Lambda}^* \in \mathrm{U}_0$ $\qquad$ $\underline{\Lambda} \in \mathrm{U}_0$

$$\int_0^T \left(\underline{\Lambda}\,\lambda^*(t) + \underline{\Lambda}^*\,\lambda(t)\right)^\top \left[\mathbb{M}\underline{\Lambda}\,\dot{\lambda}(t) + \mathbb{K}\underline{\Lambda}\,\lambda(t) - \underline{Q}(t)\right] \mathrm{dt} = 0$$

2. Fixed-point equations

$$\begin{cases} \alpha\dot{\lambda} + \beta\lambda(t) = \gamma(t) & : \text{ Ordinary Differential Equation (ODE)} \\ \underline{\Lambda}^{*\top}\left(\mathbb{J}\,\underline{\Lambda} - \underline{q}\right) = 0 & : \text{ linear Finite Element (FE) problem} \end{cases}$$

Find:

$$\alpha = ? \;\; \beta = ? \;\; \gamma(t) = ? \;\; \mathbb{J} = ? \;\; \underline{q} = ?$$

II- Implementing the PGD solver

$$\underline{\underline{W}} = \begin{pmatrix} | & & | & & | \\ \underline{W}(t_1) & \cdots & \underline{W}(t_i) & \cdots & \underline{W}(t_{N_t}) \\ | & & | & & | \end{pmatrix} \quad N_x \quad N_t$$

$$\underline{\lambda} = \left[\text{———} \right]_1 \quad N_t$$

$$\underline{\Lambda} = \begin{pmatrix} | \\ \\ | \end{pmatrix}_1 \quad N_x$$

$$\underline{W}(t) = \sum_{i=1}^{m} \underline{\Lambda}_i \, \lambda_i(t) \qquad \underline{\underline{W}} = \sum_{i=1}^{m} \underline{\Lambda}_i \, \underline{\lambda}_i = \underline{\underline{\Lambda}} \, \underline{\underline{\lambda}}$$

$$\underline{\underline{\Lambda}} = \begin{pmatrix} | & & | & & | \\ \underline{\Lambda}_1 & \cdots & \underline{\Lambda}_i & \cdots & \underline{\Lambda}_m \\ | & & | & & | \end{pmatrix}$$

Add a new mode in Matlab?

LAMBDA = [LAMBDA , Space_mode]

$$\underline{\underline{\lambda}} = \begin{pmatrix} \text{———} \; \underline{\lambda}_1 \; \text{———} \\ \vdots \\ \text{———} \; \underline{\lambda}_i \; \text{———} \\ \vdots \\ \text{———} \; \underline{\lambda}_m \; \text{———} \end{pmatrix}$$

lambda = [lambda ; Time_mode]